

Direction 2172100
Revision 0

Signa[®] Advantage 0.5T Quadrature CTL Phased Array Coil

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Operating Documentation

Language Policy For Service Documentation (Dir. 2128126)

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- ESTE MANUAL DE SERVICIO SÓLO EXISTE EN INGLÉS.
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- NÃO TENHA TENTADO REPARAR O EQUIPAMENTO SEM TER CONSULTADO E COMPREENDIDO ESTE MANUAL DE ASSISTÊNCIA TÉCNICA.
- O NÃO CUMPRIMENTO DESTES AVISOS PODE POR EM PERIGO A SEGURANÇA DO TÉCNICO, OPERADOR OU PACIENTE DEVIDO A CHOQUES ELÉTRICOS, MECÂNICOS OU OUTROS.

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- IL PRESENTE MANUALE DI MANUTENZIONE È DISPONIBILE SOLTANTO IN INGLESE.
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- SI PROCEDA ALLA MANUTENZIONE DELL'APPARECCHIATURA SOLO DOPO AVER CONSULTATO IL PRESENTE MANUALE ED AVERNE COMPRESO IL CONTENUTO.
- NON TENERE CONTO DELLA PRESENTE AVVERTENZA POTREBBE FAR COMPIERE OPERAZIONI DA CUI DERIVINO LESIONI ALL'ADDETTO ALLA MANUTENZIONE, ALL'UTILIZZATORE ED AL PAZIENTE PER FOLGORAZIONE ELETTRICA, PER URTI MECCANICI OD ALTRI RISCHI.

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- 忽略本注意事项会对维修员，操作员或病人造成触电，机械伤害或其他伤害。

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Section 1 - Introduction

1-1 How the Signa Advantage 0.5T Quad CTL Phased Array Coil Operates

The Signa Quad CTL Phased Array Coil is designed for cervical, thoracic, and lumbar spine imaging on Signa 0.5T systems with Phased Array capability. It is a receive-only Phased Array Coil containing eight coil elements and using two or four simultaneous receiver channels. The Quad Phased Array Coil uses eight signal inputs, driving eight Phased Array Input Ports. See [Illustration 1-1](#). For installation, see [Section 2-3, *Installation of Multi-Coil Preamp Assemblies for Channels 1 and 8*](#).

T/L Section: The T/L portion of the coil consists of three pairs of quad planar coil elements, each similar to the Quad T/L Spine Coil with Positioner, but smaller in the S-I direction. The two quad mode outputs of each coil element each drive a separate input port; total of six outputs. A maximum of four inputs, representing two quad coils can be selected at one time. Isolation of signal sources for the two mode outputs from each quad coil pair is accomplished as in a conventional quad planar coil. The three quad coil pairs are then isolated from adjacent coil pairs by positioning the overlap of the adjacent coils as to cancel the coupling between them. With the use of a separate preamplifier for each coil element of each quad coil pair, it is possible to fully employ the decoupling concept utilized with other Phased array coils. This ensures that optimum coil isolation is achieved from every coil at every location.

1-1 How the Signa Advantage 0.5T Quad CTL Phased Array Coil Operates (cont'd)

As in the Quad T/L Spine Coil with Positioner, each T/L Spine coil elements consists of a single loop vertical field coil, and two loop horizontal field coil. To resonate each coil, distributed capacitors about the loops are used. At the signal output port of each mode of each element of each quad coil pair, a lattice balun having an electrical length of 90 degrees is used to provide isolation of cable currents.

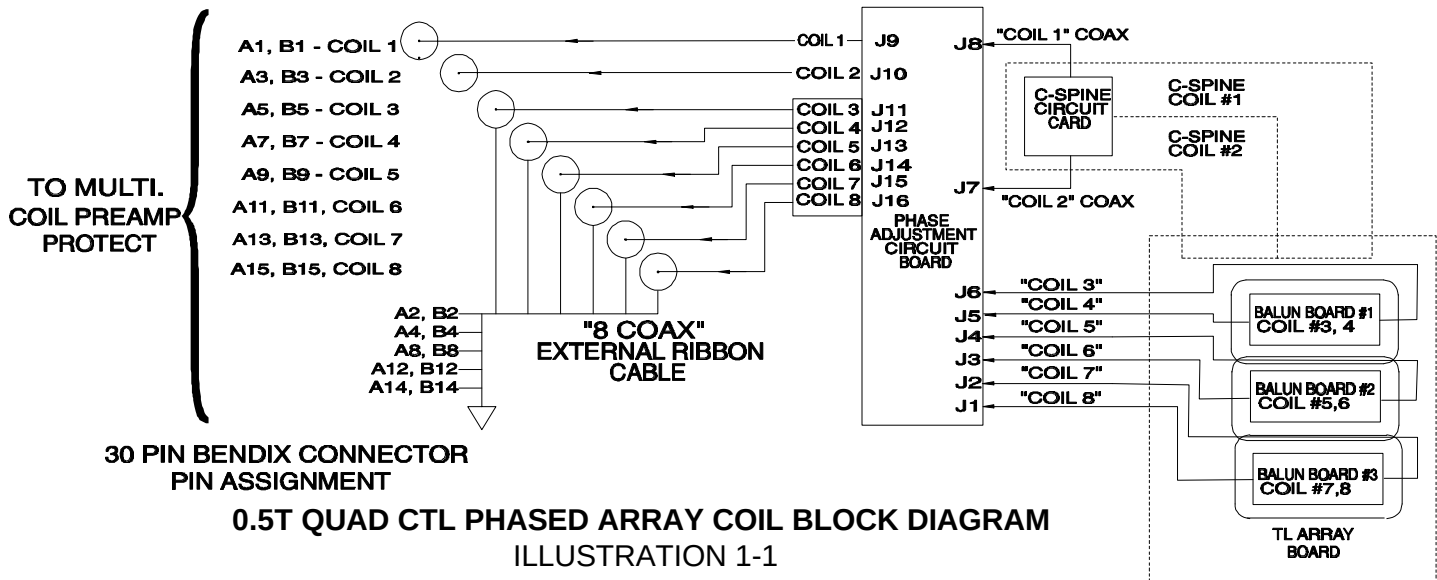
C-Spine Section: The C-Spine portion of the coil consists of two single loop conformal coil elements in the most superior region. The two superior loops are similar to the Quad C-Spine Coil, but smaller in the S-I direction. The outputs of the two coil elements drive two separate input ports, to a total of two upper cervical spine inputs. There are eight inputs; up to four coils can be selected at one time. Isolation of signal sources for the two superior coil elements is accomplished like a linear array coil. With the single-element nature of the two coil elements, the preamplifier decoupling used with other Phased Array coils can be used.

1-1 How the Signa Advantage 0.5T Quad CTL Phased Array Coil Operates (cont'd)

Transmit Decoupling: These coils are receive-only coils. Decoupling from the transmit RF field is needed. In the Horizontal Field Coils and C-Spine Coils, this is accomplished by a passive decoupling circuit on each loop of each coil element. The decoupling network consists of a pair of high speed switching diodes in a crossed configuration in series with an inductor; these components are shunted across a series distributed capacitor in each loop. The reactance of the inductor is chosen to parallel resonate the capacitor at the operating frequency of the coil. This places a high impedance in series with each loop at the capacitor location during transmit RF pulses, thus decoupling the coil from the transmit RF field. For the Vertical Field Coils, PIN diodes are used at the feedpoints for a similar function; these diodes are biased by the system T/R Driver via the coil output cable. The external cable includes eight mating SMB plugs to connect to the phase compensation board; the cable is attached by two screws to the coil top housing. A 30 pin Bendix connector is used to mate with the system Phased Array Coil Port. The Signa 0.5T Quad CTL Phased Array Coil drives eight coil inputs, Ports 1, 2, 3, 4, 5, 6, 7, 8 - corresponding to receiver channels 0, 1, 2, 3, 0, 1, 2, 3 respectively.

Pin diodes for the Horizontal coil are for T/R Driver Fault Detection only. They perform no decoupling function.

1-1 How the Signa Advantage 0.5T Quad CTL Phased Array Coil Operates (cont'd)



1-2 Compatibility

The 0.5T Signa advantage Quadrature CTL Phased Array Coil is compatible with the following hardware configurations:

- Signa® Advantage™ 0.5T System with Phased Array upgrade

1-3 Related Documents

- Direction 15400, Signa Advantage 1.5T, 1.0T, & 0.5T System
- Direction 2124201-3, MR release 5.x Signa Service Methods

1-4 Organization of this Document

This manual is divided into the following sections:

Section 1, *Introduction*, describes how the 0.5T Quad CTL Phased Array Coil operates.

Section 2, *Setup and Calibration*, describes installation procedures.

Section 3, *Functional Checks*, describes the normal power-up sequence.

Section 4, *Replacement/Maintenance*, describes field maintenance procedures.

Section 5, *Renewal Parts*, lists field replaceable parts.

1-5 Environmental Requirements

Operate and store the 0.5T Quadrature CTL Phased Array Coil in the Scanner Room.

Dimensions: 126 cm x 42.7 cm x 17.5 cm (49.6" x 16.8" x 6.9")

Section 2 - Setup and Calibration

2-1 Checking the Shipping List

Table 2-1 lists the M1687CT Signa Advantage 0.5T Quadrature CTL Phased Array Coil parts. Check that all parts have been shipped.

TABLE 2-1
0.5T QUADRATURE CTL PHASED ARRAY COIL SHIPPING LIST

<u>QTY.</u>	<u>ITEM</u>	<u>PART NUMBER</u>
1	0.5T Quadrature CTL Phased Array Coil	2151178-2
2	0.5T Multicoil Preamp Asm.	46-328040G1
1	Phantom Holder	2100937-21
4	100 mm Phantoms	46-317586G1
1	Operator Manual	2172124-100
1	Service Manual	2172100
1	Forehead Strap	E 8801 KF
1	Chin Strap	E 8801 KE

2-2 Installing the Signa Advantage 0.5T Quadrature CTL Phased Array Coil

At the Console, install a new softkey. This softkey will be used by the operator to select CTLTOP, CTLBOT, CTLMID, CS12, TS34, TS56, and LS78.

Refer to the Signa & Signa Advantage System Manuals for information on installing soft keys (use Coil/Phased Array default values in Table 2-2 and [Table 2-3](#)).

TABLE 2-2
COIL VALUES

SYSTEM	COIL NAME	COIL TYPE	EXTREMITY COIL	CABLE LOSS	COIL LOSS	RECON FACTOR	SCALE (RSF)	XMIT ATTN	XMIT COIL	MULTI-COIL
0.5T	CTLTOP thru LS78	SURF.	NO	1.23	0.076 +	Head Factor x 1		0	QUAD	YES
				NUMBER OF <u>FAST RECEIVERS</u>	START FAST <u>RECEIVER</u>					
				0	4					
						STOP FAST <u>RECEIVER</u>				
						4				

2-2 Installing the 0.5T Quadrature CTL Phased Array Coil (cont'd)

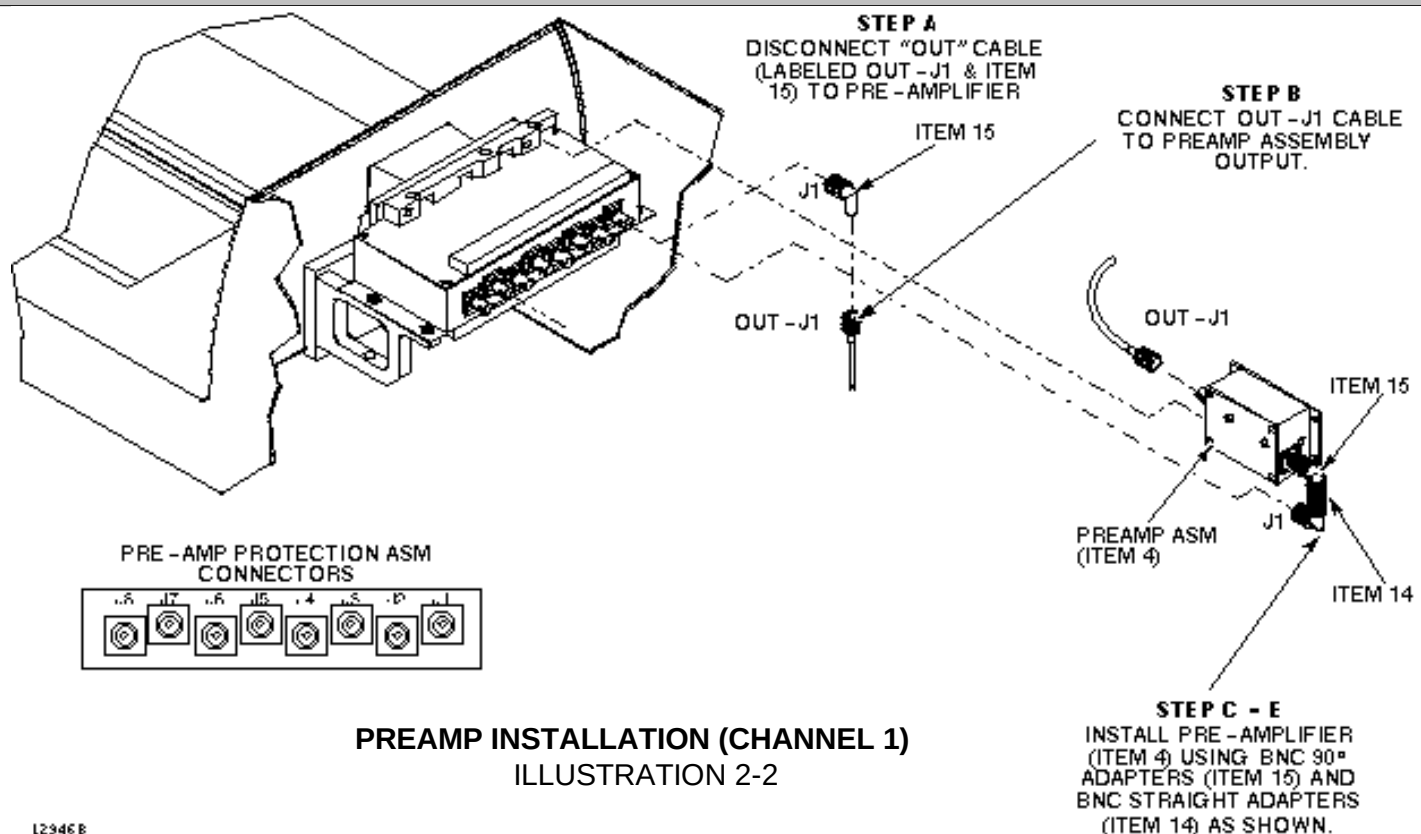
TABLE 2-3
PHASED ARRAY VALUES

COIL NAME	NUMBER OF RECEIVERS	START RECEIVER	STOP RECEIVER	PORT ENABLE MASK	ERROR ENABLE MASK
CTLTOP	4	0	3	3	3
CTLBOT	4	0	3	12	12
CTLMID	4	0	3	6	6
CS 12	2	0	1	1	1
TS 34	2	2	3	2	2
LS 56	2	0	1	4	4
LS 78	2	2	3	8	8

NOTE: If the customer has an M1687AB Linear Array and wishes to retain the option, then add a 'Q' prefix to all Quad CTL names.

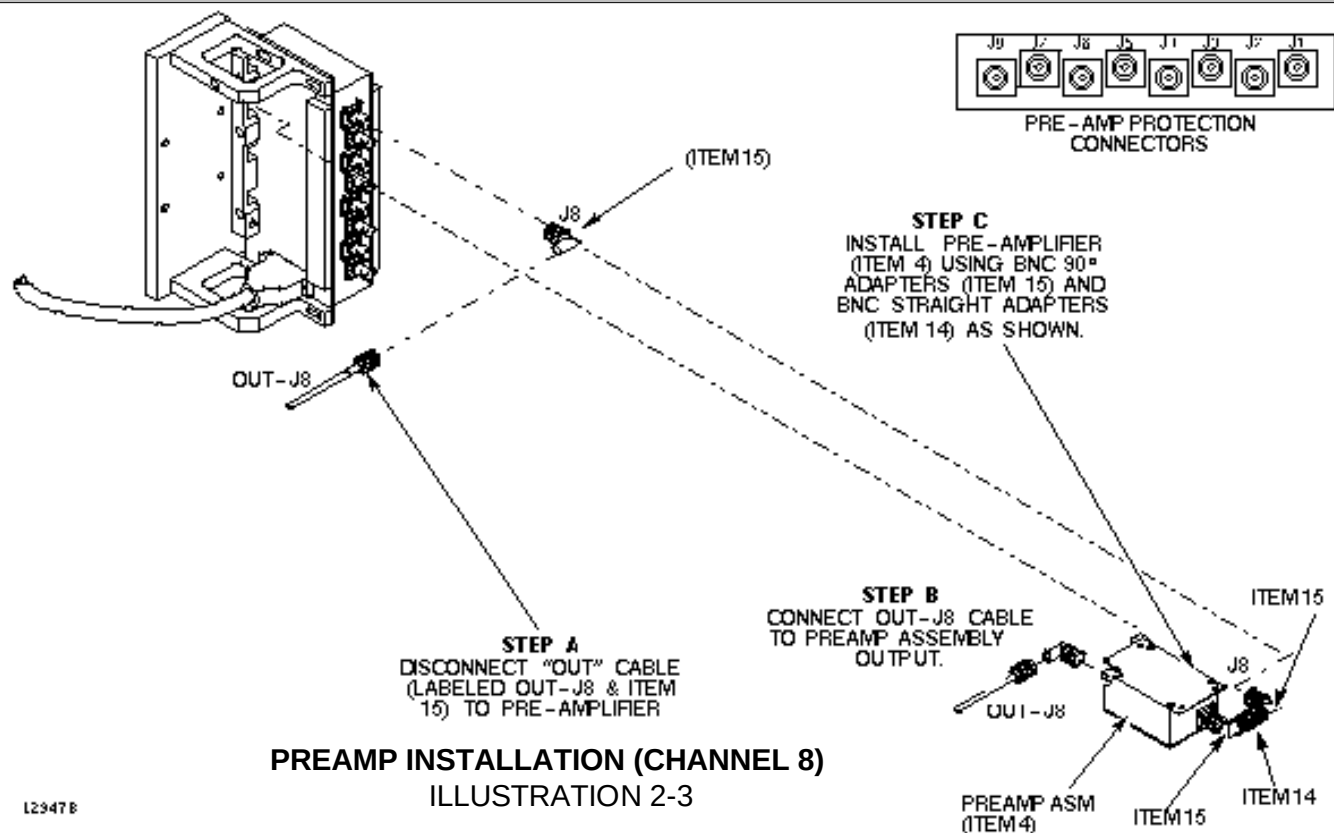
2-3 Installation of Multi-coil Preamp Assemblies for Channel 1 & 8

1. Remove Carriage Cover Access Panel from rear of Carriage Cover.
2. Install Channel 1 upper preamplifier assembly, per [Illustration 2-2](#).
 - a. Disconnect output cable J1 and BNC right angle adapter (item 15) from channel 1 OUTPUT of preamp Protection Assembly. Discard BNC right angle adapter.
 - b. Connect output cable J1 to new preamplifier OUTPUT.
 - c. Insert new Preamplifier assembly into applicable notch on Preamp Protection locating block.
 - d. Assemble two BNC right angle connectors and straight adaptors to new Preamplifier INPUT.
 - e. Connect right angle adapter to J1 on Preamp Protection assembly from Preamplifier INPUT.



2-3 Installation of Multi-coil Preamp Assemblies for Channel 1 & 8 (cont'd)

3. Install Channel 8 lower preamplifier assembly, see [Illustration 2-3](#).
 - a. Disconnect output cable J8 and BNC right angle adapter (item 15) from channel 8 OUTPUT of Preamp Protection Assembly. Connect output cable J8 and BNC right angle adapter to new preamplifier OUTPUT.
 - b. Insert new Preamplifier assembly into applicable notch on Preamp Protection locating block.
 - c. Assemble two BNC right angle connectors and straight adaptors to new Preamplifier INPUT. Connect right angle adapter to J8 on Preamp Protection assembly from Preamplifier INPUT.



2-4 Functional Checks

1. Perform a Body Coil Scan Performance Verification. Refer to [Section 3-1, Signal Advantage 0.5T Scanner Verification](#).
2. Perform a Quad CTL Phased Array Coil Performance Verification. Refer to [Section 3-2, Quadrature CTL Phased Array Coil Imaging Performance Verification](#).

2-5 Periodic Quality Assurance Check

On a periodic basis, such as during planned maintenance, perform the quality assurance checks outlined below to ensure that the coil is operating properly:

1. Check the external cable for cracks or breaks once a week. Refer to [Section 3-4, Checking the Protection PIN Diodes and Output cable with DMM](#).
2. Perform a coil SNR verification. Refer to [Section 3-2, Quad CTL Phased Array Coil Imaging Performance Verification](#).
3. Record the date and value calculated in Section 3-3, SNR Image Analysis in column 2 or 3 under "SNR Data QA Check" of the [Data Table](#) as is instructed.
4. As is instructed in the Data Table, divide the SNR value obtained in the periodic QA check by the original SNR value and record in column 4 of the [Data Table](#).
5. If this ratio is not greater than 85%, then there may be a problem in the Multicoil system. Contact your local GE Service Representative.

Section 3 - Functional Checks

3-1 Signa Advantage 0.5T Scanner Verification

Note

An alternate proprietary procedure is available for GE use and to customers with a valid Advanced Service Package Limited License. Refer to "TLT PROCEDURE" located on *CD-ROM Direction 15520, navigate to System: Troubleshooting.*

Phantom Required

4 -100 mm Phantoms, 46-317586G1

1 - Phantom Holder, 2100937-21

3-1 Signa Advantage 0.5T Scanner Verification (cont'd)

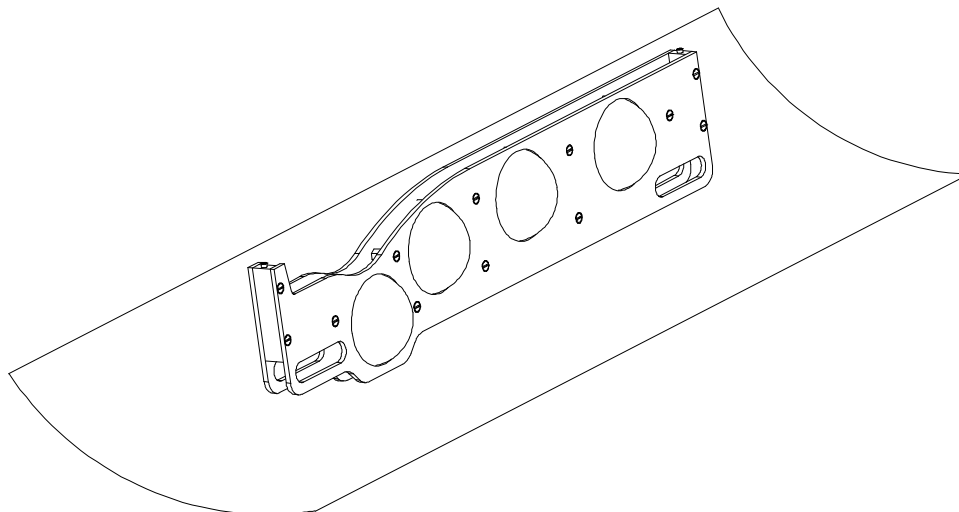
Setup Procedure

Caution

The Quad Head Coil must be completely removed from the cradle before performing any body scans. Failure to do this may result in damage to the Head Coil T/R Network.

1. Remove Quad Head Coil (if present) from cradle.
2. Select **[New Series]** to allow a new Landmark to be set.
3. Position the CTL Phantom/Positioner with phantoms in the center of the cradle. Place the phantom/positioner upside-down on the table. Landmark the center of the positioner with phantoms and advance to isocenter using the **[ADVANCE TO SCAN]** button. See Illustration 3-1.

3-1 Signa Advantage 0.5T Scanner Verification (cont'd)



QUAD CTL PHANTOM/POSITIONER LANDMARK SETUP
ILLUSTRATION 3-1

3-1 Signa Advantage 0.5T Scanner Verification (cont'd)

4. Setup Scan Prescription as shown in Table 3-1.

TABLE 3-1
BODY COIL SNR - SCAN PROTOCOL

Id:	geservice	Rep Time (TR):	[300 msec]
Name:	body snr	Auto CF:	[Peak]
Patient Weight:	300	Field of View:	[40 cm]
Patient Entry:	[Head First]	Scan Thickness:	[5 mm]
Patient Position:	[Supine]	Interscan Spacing:	[Other] 0
Axial/Sag. Landmark:	[Sternal Notch]	Start Loc (L/R):	0
Coil Type:	[Body Coil]	End Loc (L/R):	0
Scan Plane:	[Sagittal]	No. of Scan Location:	1
Image Mode:	[2D]	FOV Center (P/A):	0 (I/S) 0
** SAR must be on **	[Monitor SAR]	Acq. Matrix (freq.):	[256]
Pulse Sequence:	[Spin Echo]	Acq. Matrix (phase):	[256]
Imaging Options:	[None]	Frequency Direction:	[S/I]
or enter PSD Filename		Imaging Time:	[1 NEX 1:23]
Number of Echoes:	[1]	Contrast:	[No]
Echo Time (TE):	[30 msec]	Table Delta:	0 mm

3-1 Signa Advantage 0.5T Scanner Verification (cont'd)

5. Select **[Auto Prescan]** to properly calibrate the RF power level for the 90 degree and 180 degree pulses.
6. Select **[Scan]**. Observe the resulting images. Ensure that there are no artifacts of any sort in the resulting image. Record the Exam number and Series number for SNR Calculations.
7. Select **[Scan]** again. This second image will be used for determination of Body Coil mode SNR.
8. Select **[Cancel]**. Refer to Section 3-3 for SNR Image Analysis.

3-2 Quadrature CTL Phased Array Coil Imaging Performance Verification

Note

An alternate proprietary procedure is available for GE use and to customers with a valid Advanced Service Package Limited License. Refer to "TLT PROCEDURE" located on *CD-ROM Direction 15520, navigate to System: Troubleshooting*.

Phantom Required

- 4 -100 mm Phantoms, 46-317586G1
- 1 - Phantom Holder, 2100937-21

3-2 Quadrature CTL Phased Array Coil Imaging Performance Verification (cont'd)

Setup Procedure

1. Select **[New Series]** to allow a new Landmark to be set.

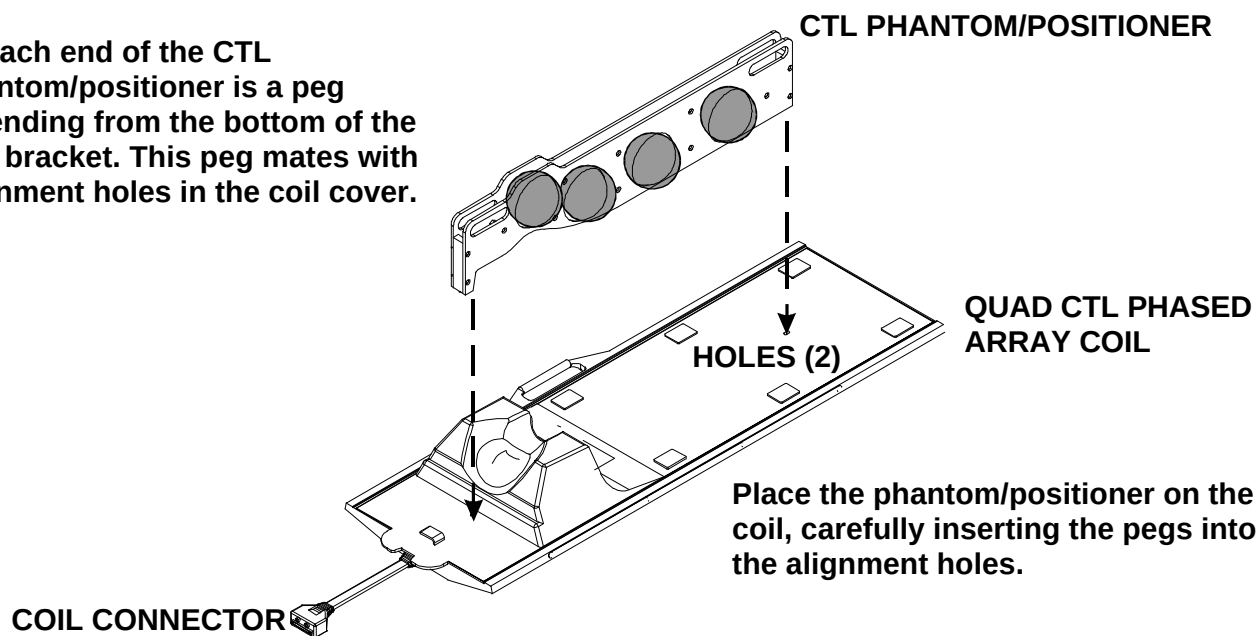
Caution

The Quad Head Coil must be completely removed from the cradle before performing any body scans. Failure to do this may result in damage to the Head Coil T/R Network.

2. Remove Quad Head Coil (if present) from cradle.
3. Place Quad CTL Phased Array Coil to be tested on table.
4. Connect Coil connector to the mating connector in Carriage Assembly.
5. Place the phantom/positioner on the Coil. Ensure the two pegs on the bottom of the positioner mate with the holes on the top cover of the Coil. See [Illustration 3-2](#).

3-2 Quadrature CTL Phased Array Coil Imaging Performance Verification (cont'd)

At each end of the CTL phantom/positioner is a peg extending from the bottom of the end bracket. This peg mates with alignment holes in the coil cover.



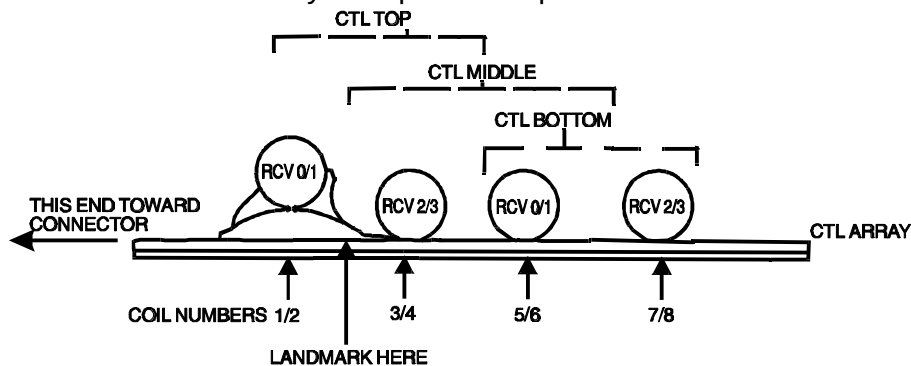
QUAD CTL PHANTOM/POSITIONER SETUP

ILLUSTRATION 3-2

3-2 Quadrature CTL Phased Array Coil Imaging Performance Verification (cont'd)

6. Position the Quad CTL Phased Array Coil and Positioner with phantoms in the center of the cradle. Landmark for center of Coil and advance to isocenter using the **[ADVANCE TO SCAN]** button. See Illustration 3-3.

Note: Testing the CTL Coil requires two scans. Landmark in the center of the coil-See Illustration 3-3. Scan the CTLTOP (S180), this tests the top half of the array. Scan the CTLBOT (I180), this tests the bottom half of the array. The phantom spheres are on 180 mm centers.



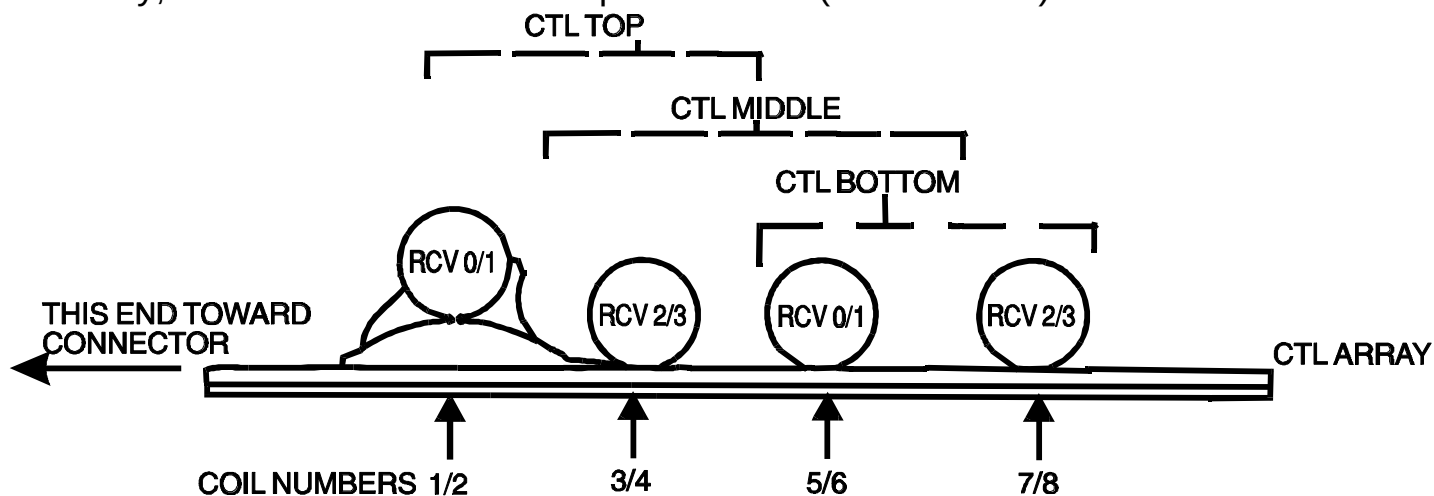
Note: The Quad CTL array contains eight coils; four are used at one time. Two regions are designated. Landmarks are marked on the side of the array. The CTLTOP array uses coils 1 through 4 - Black Marking. CTLBOT uses coils 5 through 8 - Red Marking. If CTL Coil is landmarked as above, Scan Location for CTLTOP is S180 and CTLBOT is I180.

PHANTOM/POSITIONER LANDMARK :
ILLUSTRATION 3-3

3-2 Quadrature CTL Phased Array Coil Imaging Performance Verification (cont'd)

Phased Array Coil Receiver Paths

The Phased Array Coils have multiple individual antenna loop elements inside the assembly; the TPS chassis has multiple receivers (RCV0-RCV3). See Illustration 3-4.



QUAD CTL PHASED ARRAY COIL TO TPS RECEIVER CORRESPONDENCE
ILLUSTRATION 3-4

3-2 Quadrature CTL Phased Array Coil Imaging Performance Verification (cont'd)

7. Setup Scan Prescription as shown in Table 3-2.

TABLE 3-2
Quadrature CTL Phased Array Coil SNR - Scan Protocol

Id:	geservice	Rep Time (TR):	[300 msec]
Name:	quad ctl snr	Auto CF:	[Peak]
Patient Weight:	300	Field of View:	[40 cm]
Patient Entry:	[Head First]	Scan Thickness:	[5 mm]
Patient Position:	[Supine]	Interscan Spacing:	[Other] 0
Axial/Sag. Landmark:	[Sternal Notch]	Start Loc (L/R):	0
Coil Type:	[Other Coils] [CTLTOP]	End Loc (L/R):	0
Scan Plane:	[Sagittal]	No. of Scan Location:	1
Image Mode:	[2D]	FOV Center (P/A):	0 (I/S): S180
** SAR must be on **	[Monitor SAR]	Acq. Matrix (freq.):	[256]
Pulse Sequence:	[Spin Echo]	Acq. Matrix (phase)	[256]
Imaging Options:	[None]	Frequency Direction:	[S/I]
or enter PSD Filename		Imaging Time:	[1 NEX 1:23]
Number of Echoes:	[1]	Contrast:	[No]
Echo Time (TE):	[30 msec]	Table Delta:	0 mm

3-2 Quadrature CTL Phased Array Coil Imaging Performance Verification (cont'd)

8. Select **[Auto Prescan]** to properly calibrate the RF power level for the 90 degree and 180 degree pulses.

9. Select **[Adv to Scan]**.

10. Select **[Scan]**. Observe the resulting image of the spheres. Ensure that there are no artifacts of any sort in the sphere image. Record the Exam number and Series number for SNR Calculations.

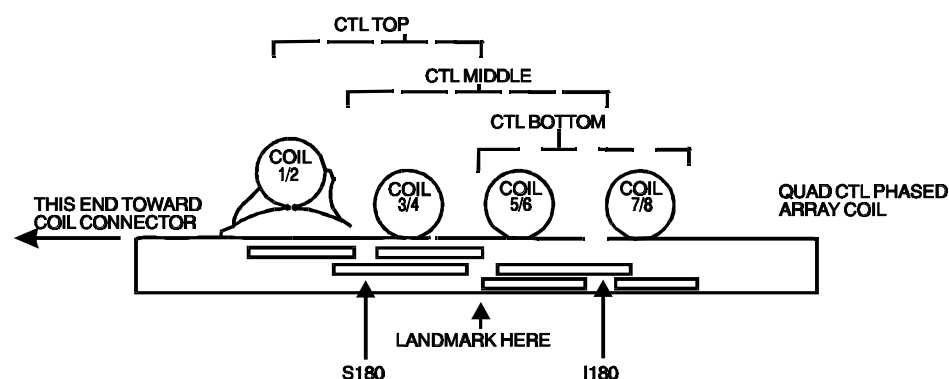
11. Select **[Scan]** again. This second image of the sphere will be used for determination of CTLTOP mode SNR.

12. Select **[Cancel]** and **[New Series]**.

3-2 Quadrature CTL Phased Array Coil Imaging Performance Verification (cont'd)

13. Ensure the landmark is still on center of coil. See Illustration 3-5.

Note: Testing the CTL Coil requires two scans. Landmark in the center of the coil-See Illustration 3-5. Scan the CTL TOP (S180), this tests the top half of the array. Scan the CTL BOT (I180), this tests the bottom half of the array. The phantom spheres are on 180 mm centers.



Note: The Quad CTL array contains eight coils; four are used at one time. Two regions are designated. Landmarks are marked on the side of the array. The CTLTOP array uses coils 1 through 4 - Black Marking. CTLBOT uses coils 5 through 8 - Red Marking. If CTL Coil is landmarked as above, Scan Location for CTLTOP is S180 and CTLBOT is I180.

PHANTOM/POSITIONER LANDMARK SETUP CTLBOT
ILLUSTRATION 3-5

3-2 Quadrature CTL Phased Array Coil Imaging Performance Verification (cont'd)

14. Select **[Patient Position]**, **[Other Coils]**, **[CTLBOT]**.
15. Select **[Scanning Range]**, FOV center in S/I direction as **I180**, and **[Accept]**. Advance to **CURRENT ACQ**.
16. Select **[Auto Prescan]** to calibrate the RF power level for the 90 and 180 degree pulses.
17. Select **[Scan]**. Observe the resulting image of the spheres. Ensure that there are no artifacts of any sort in the sphere images. Record the Exam number and Series number for SNR calculations.
18. Select **[Scan]**. This second image of the spheres will be used for determination of CTLBOT mode SNR.
19. Select **[Cancel]**. Refer to [Section 3-3](#) for SNR image analysis.

3-3 SNR Image Analysis

Description

The SNR tool retrieves two operator selected images. Signal value is computed as the mean pixel value in a ROI covering 80% of the image. The image is analyzed to determine the center of the image for positioning the ROI. A difference image is created by subtracting the second image from the first and the same ROI is used to calculate noise from the subtracted image. The signal value, noise value, and signal to noise ratio are reported. There is an option to save the difference image with the results annotated.

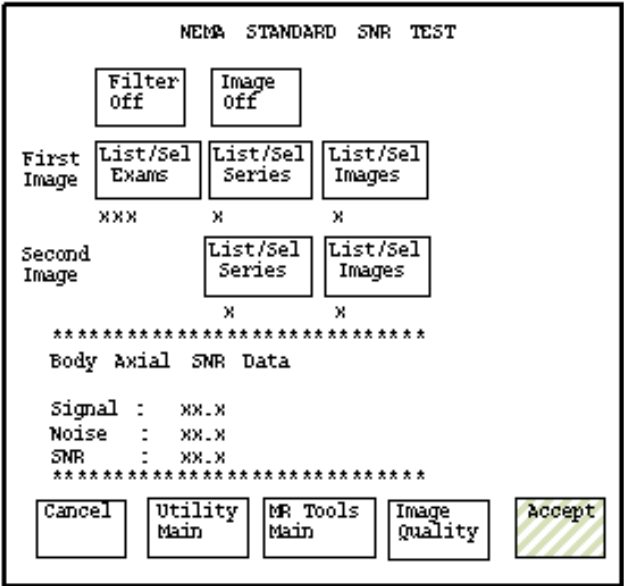
3-3 SNR Image Analysis (cont'd)

Procedure

1. Touch **[UTILITIES]**, **[MR Tools]**, **[Image Quality]**, then **[SNR Test]**. The SNR Test screen is displayed; see Illustration 3-6.
2. Enter image exam, series, and image numbers. If exam, series, or image numbers are not known, select **[List Exams]**, **[List Series]**, or **[List Images]** to display list to choose from.

Note

Image number selection must be back lit (highlighted) to be able to enter information. Use Switch key on keyboard to transfer control from left to right side of Touch Screen.



Note: Accept changes to continue only after an analysis has been performed.

SNR TEST SCREEN
ILLUSTRATION 3-6

3-3 SNR Image Analysis (cont'd)

3. If high pass filtering is desired to be performed on data, touch **[Filter Off]** which will highlight and change to **[Filter On]**.
4. If the difference image annotated with data is to be created, touch **[Image Off]** which will highlight and change to **[Image On]**.
5. Touch **[Accept]** to begin analysis. The final values are displayed on the touch screen, see Illustration 3-6.
6. Touch **[Continue]** then select the next exam and repeat the above analysis for each image pair.
7. Record the date and value calculated in the appropriate column under "SNR Data QA Check" of the Data Table as is instructed.

3-4 Checking the Protection PIN Diodes and Output Cable with DMM

CAUTION

As coil tuning and decoupling tuning may be affected by PIN diode replacement, field replacement of the PIN diodes will not be performed.

1. Select the DIODE TEST function on the DMM.
2. Connect the NEGATIVE lead to the the connector pin on row A. Connect the POSITIVE lead to the pin on row B. Refer to Table 3-3 and Illustration 3-7. A reading of 0.500 volts to 0.900 volts should be observed.
3. If a reading below 0.400 volts is observed in either direction, either the output cable is shorted or the PIN diode is defective.
4. If a reading above 0.900 is noted, the output cable or the PIN diode is open.

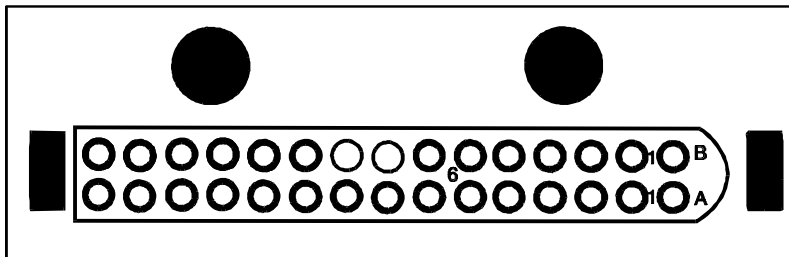
3-4 Checking the Protection PIN Diodes and Output Cable with DMM (cont'd)

5. Connect the NEGATIVE lead to the connector pin on row B. Connect the POSITIVE lead to the connector pin on row A. Refer to Table 3-3.
6. A reading of INFINITY should be observed. If a finite reading is observed in both directions, the output cable or the PIN diodes are shorted or leaky.
7. To test the cable, perform the above tests while flexing the cable, especially near the 30 pin connector and strain relief exiting the coil.

3-4 Checking the Protection PIN Diodes and Output Cable with DMM (cont'd)

TABLE 3-3
Diode Test Connections

COIL NUMBER	POSITIVE LEAD CONNECTION		NEGATIVE LEAD CONNECTION	
	STEP 2	STEP 6	STEP 2	STEP 6
1	1A	4A	4B	1B
2	3A	4A	4B	3B
3	5A	4A	4B	5B
4	7A	8A	8B	7B
5	9A	8A	8B	9B
6	11A	12A	12B	11B
7	13A	12A	12B	13B
8	15A	14A	14B	15B



PIN LOCATIONS FOR PHASED ARRAY CONNECTOR
ILLUSTRATION 3-7

3-5 Checking the Decoupling Diodes with a DMM

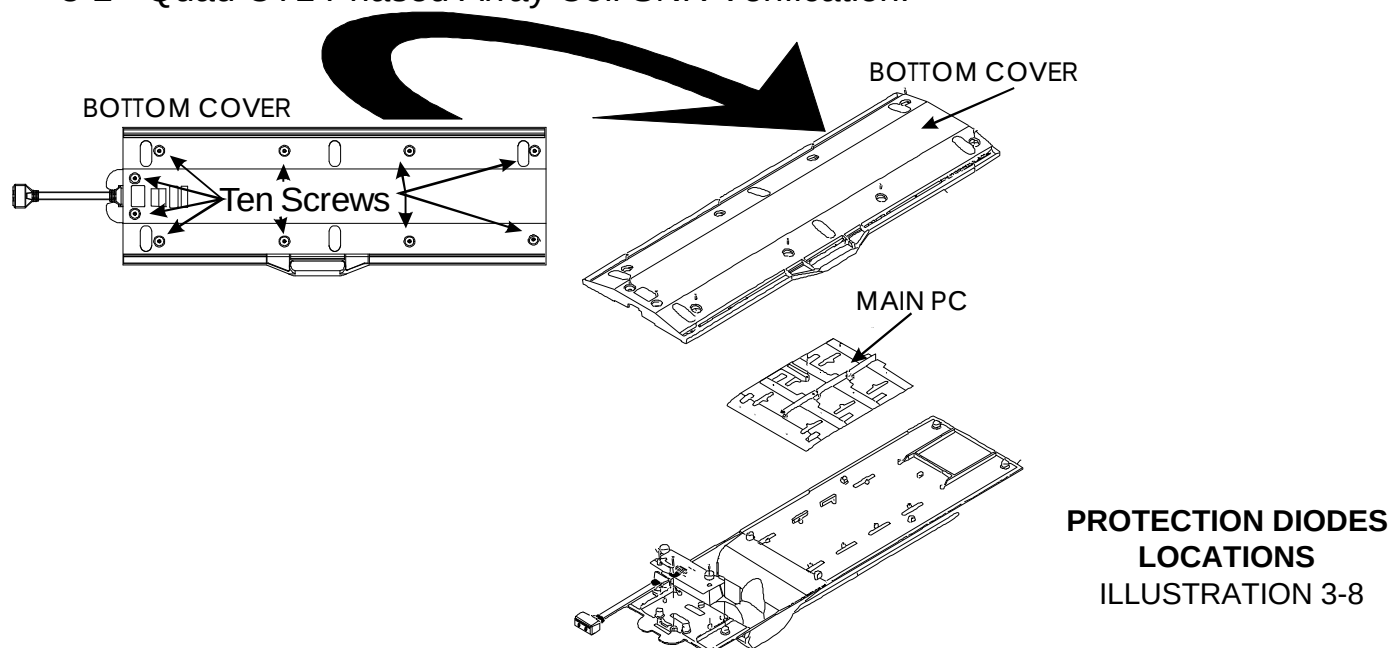
CAUTION

As coil tuning and decoupling tuning may be affected by decoupling diode replacement, field replacement of the decoupling diodes will not be performed.

1. Remove the bottom cover of the coil. [See Section 4-1](#).
2. Locate the 8 sets of decoupling diodes; two on each of the three T/L PC boards, and one set in each loop of the superior C-Spine PC board. [See Illustration 3-8](#).
3. Measure across each pair of diodes in both directions with a DMM set to the diode test function. The reading should be 0.400 volts to 0.900 volts in each direction. A lower reading indicates a shorted or leaky diode. A higher reading indicates an open diode.
4. If any of the diode pairs have failed, the coil should be replaced.

3-5 Checking the Decoupling Diodes with a DMM (cont'd)

5. Attach the Bottom Cover. [See Section 4-1](#). Perform the performance tests in Section 3-2 - Quad CTL Phased Array Coil SNR Verification.



Section 4 - Replacement / Maintenance

4-1 Disassembly / Assembly of 0.5T Quad CTL Phased Array Coil

Note:

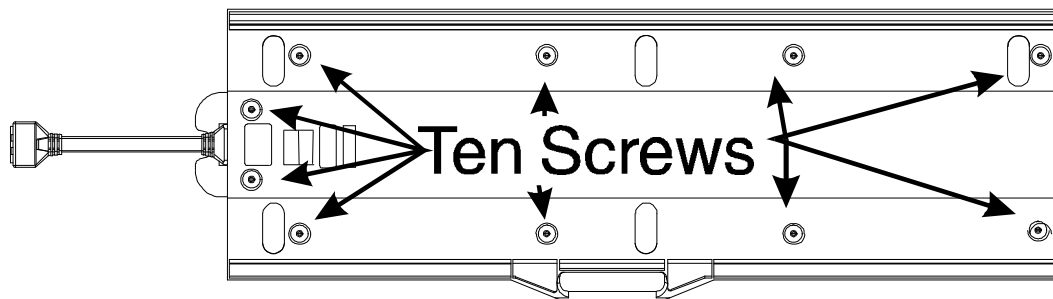
Each antenna loop element is tuned at the factory to provide the proper input impedance. None of these boards are field replaceable. These circuits must remain together as a matched set.

1. Remove the ten screws holding the top and bottom housing together. See Illustration 4-1.
2. Separate the two housings from each other.
3. To reassemble, put the two housings together and **lightly** tighten the ten screws.

Note:

The screws are very susceptible to breakage from overtightening.

4-1 Disassembly / Assembly of 0.5T Quad CTL Phased Array Coil (cont'd)

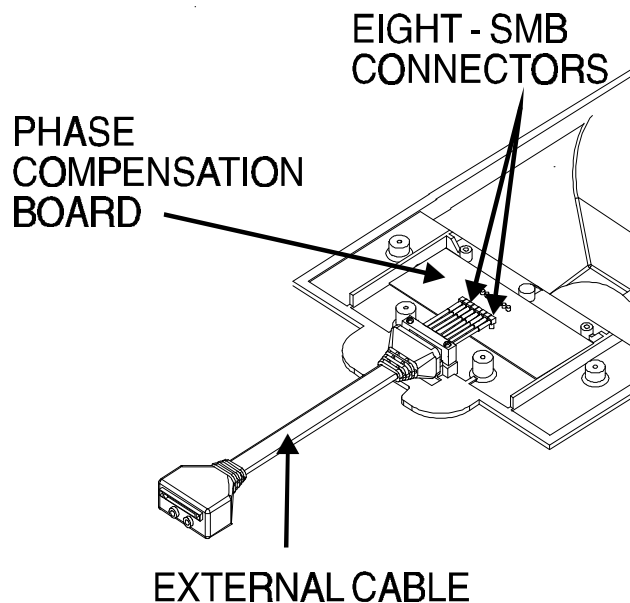


DISASSEMBLY/ASSEMBLY OF QUAD CTL PHASED ARRAY COIL
ILLUSTRATION 4-1

4-2 Replacing the External Cable.

1. Disassemble the top and bottom housings. [Refer to Section 4-1](#).
2. Remove the eight SMB connectors on the External Cable from the mating connectors on the Phase Compensation Board. Remove the two screws and the Strain Relief holding the External Cable in place. Remove the External Cable.
3. Install the new External Cable in place, ensuring that the Strain Relief is placed in the same direction as the original. [See Illustration 4-2](#).
4. Install the Strain Relief and screws in place.
5. Reconnect the eight SMB connectors to the Phase Adjustment Board in the proper order.
6. Assemble the top and bottom housings together. [Refer to Section 4-1](#).
7. Repeat the [Periodic Quality Assurance Check in Section 2-4](#).

4-2 Replacing the External Cable. (cont'd)



REPLACEMENT OF EXTERNAL CABLE
ILLUSTRATION 4-2

4-3 Replacing the Protection (PIN) Diodes

CAUTION

As coil tuning and decoupling tuning may be affected by PIN diode replacement, field replacement of the PIN diode will not be performed.

1. PIN Diodes are not field replaceable. Return the coil for replacement.

4-4 Cleaning the Coil

Caution

Avoid damaging sensitive electronic parts. Do not spray or pour dishwashing solution directly onto the Quadrature CTL Phased Array Coil, or external cable. Do not use alcohol to clean the Quadrature CTL Phased Array Coil. Never submerge the Quadrature CTL Phased Array Coil in any liquid.

Clean the Quadrature CTL Phased Array Coil and external cable with a mild dishwashing liquid and water solution. Wet a soft cloth with the solution and proceed to clean.

Before returning the coil for servicing, thoroughly clean the coil, cable, and connectors with a 10% bleach in water solution.

Section 5 - Renewal Parts

WARNING

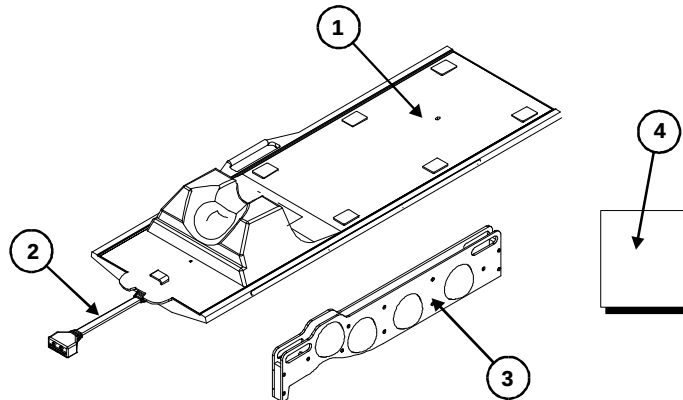
BIO HAZARD! EQUIPMENT BEING RETURNED FROM USE IN A CLINICAL SETTING MUST BE CLEAN AND FREE OF BLOOD AND OTHER INFECTIOUS SUBSTANCES. THE DEPARTMENT OF TRANSPORTATION (DOT) HAS RULED THAT ITEMS THAT WERE SATURATED AND/OR DRIPPING WITH HUMAN BLOOD THAT ARE NOW CAKED WITH DRIED BLOOD; OR WHICH WERE USED OR INTENDED FOR USE IN PATIENT CARE ARE REGULATED MEDICAL WASTE FOR TRANSPORTATION PURPOSES AND MUST BE TRANSPORTED AS A HAZARDOUS MATERIAL. UNDER NO CIRCUMSTANCES SHOULD A PART OR EQUIPMENT WITH VISIBLE BODY FLUIDS BE TAKEN OR SHIPPED FROM A CLINIC OR SITE (FOR EXAMPLE, SURFACE COILS).

1. Employees shall follow proper decontamination procedures for clean up of bloodborne pathogens. Refer to Section 4-4, Cleaning The Coil. It is the responsibility of the GEMS employee to insure the part/equipment has been properly decontaminated prior to shipment.

Section 5 - Renewal Parts - cont'd

0.5T Signa Advantage Quadrature CTL Phased Array Coil

2151177



<u>Item</u>	<u>Part Number</u>	<u>FRU</u>	<u>Name</u>	<u>Qty.</u>	<u>Description (Remarks)</u>
1	2151178-2	1	Quad CTL	1	0.5T Advantage Quad CTL Phased Array Coil w/Accessories
2	2100937-15	1	Cable	1	External Cable
3	2100937-21	1	Holder	1	0.5T Quad CTL Phantom Holder
4	46-317586G1	2	Phantom	4	100 mm TLT Phantoms
5	46-288292G1	1	Circuit Board	1	0.5T Preamplifier

Data Table

Use the space provided below to record the calculated signal to noise ratio (SNR) data obtained from Section 3, Functional Checks. After recording the SNR data obtained during the initial coil installation, record subsequent SNR data in column 2 - 3 below and the date they were obtained in column number 1 as a periodic QA check. If the ratio of any of the coils found is not greater than 85%, then there is a problem in the coil or the MR system.

Original SNR data obtained at initial coil installation:

Body Coil SNR Value: _____

Quadrature CTL Phased Array Coil SNR Value: CTLTOP _____
 CTLBOT _____

Date: _____

Data Table (continued)

SNR Data QA (Quality Assurance) Check:

1	2	3	4
<u>Date:</u>	<u>CTLTOP</u>	<u>CTLBOT</u>	<u>Are new Values Divided by Original Values > 85%?</u>
_____	_____	_____	_____ Y/N
_____	_____	_____	_____ Y/N
_____	_____	_____	_____ Y/N
_____	_____	_____	_____ Y/N
_____	_____	_____	_____ Y/N
_____	_____	_____	_____ Y/N
_____	_____	_____	_____ Y/N
_____	_____	_____	_____ Y/N
_____	_____	_____	_____ Y/N
_____	_____	_____	_____ Y/N
_____	_____	_____	_____ Y/N
_____	_____	_____	_____ Y/N

Make additional copies of this document as is needed.

Defective Coil Return Form

Note

To allow for proper assessment of defective returned coils, this form must be completely filled out and accompany all returned coils. Include films or prints of any image quality related complaints with a description of scan protocol used.

Date _____
Site Name _____
Site Address _____
Service Engineer _____
Coil Serial Number _____
Date Coil Installed _____
Description of Coil Problem _____

Defective Coil Return Form (continued)

Note

An alternate proprietary procedure is available for GE use and to customers with a valid Advanced Service Package Limited License. Refer to "TLT PROCEDURE" located on *CD-ROM Direction 15520, navigate to System: Troubleshooting*. Use Table DR-1 and DR-2 to record the TLT results of the defective coil.

TABLE DR-1
TLT DATA FOR DEFECTIVE SURFACE COIL

SITE: _____ NAME: _____ DATE: _____
TLT FILE NUMBER: _____
SNR: NOISE _____
SNR MEAN _____
SNR SIGNAL _____
SNR AREA _____
TR MAP: 89-91 _____ %
85-95 _____ %
65-115 _____ %
FLIP MEAN _____

Defective Coil Return Form (continued)

TABLE DR-2
TLT DATA FOR DEFECTIVE PHASED ARRAY COIL

SITE: _____ NAME: _____ DATE: _____

TLT FILE NUMBER: _____

SNR: NOISE (0) _____

NOISE (1) _____

NOISE (2) _____

NOISE (3) _____

NOISE (C) _____

SNR MEAN (0) _____

SNR MEAN (1) _____

SNR MEAN (2) _____

SNR MEAN (3) _____

SNR MEAN (C) _____

TR MAP: 89-91 (C) _____ %

(Combined 85-95 (C) _____ %

Results) 65-115 (C) _____ %

FLIP MEAN (C) _____

FLIP SDV (C) _____

RCV MEAN (C) _____

RCV SDV (C) _____

Damage in Transportation Statement

All packages should be closely examined at time of delivery. If damage is apparent, have notation **"damage in shipment"** written on **all** copies of the freight or express bill **before** delivery is accepted or "signed for" by a General Electric representative or a hospital receiving agent. Whether noted or concealed, damage **MUST** be reported to the carrier **immediately** upon discovery, or in any event, within **14** days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this **14** day period.

Call Traffic and Transportation, Milwaukee, WI (414) 785-5052/8*323-5052 **immediately** after damage is found. At this time be ready to supply name of carrier, delivery date, consignee name, freight or express bill number, item damaged and extent of damage.

Complete instructions regarding claim procedure are found in Section "S" of the Policy & Procedure Bulletins.

3/12/92



GE Medical Systems

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(Asia, Pacific, Latin America, North America)

GE Medical Systems — Europe: Telex 261794
Shortlands, Hammersmith, London W6 8BX U.K.

Note Page (optional)

NOTES -- This page is used to keep all of the notes for this procedure.
(Optional)

↑

↓

Release Notes (required)

RELEASE NOTES -- This page is used to hold all release notes for this procedure (required)

This is the first release of this procedure in Toolbook. There are no notes.

Initial Comments (required)

INITIAL COMMENTS -- This page is used to hold all initial comments for a procedure. (Required)

Revision 0 December, 1996 Initial Release.

Revision Information (required)

REVISION INFORMATION - this is for revision information for a procedure.
(Required)

Revision#: 0

Date: 12/12/96

Class: A

87305-T-401

Author: George E. Lang

Title: Signa® Advantage 0.5T Quadrature CTL Phased Array Coil

Revision Description:



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